pedicellate flowers at the bottom. All flowers appear at the same level to form convex or flat topped racemose inflorescence. Example: *Caesalpinia*. **Compound corymb:** A branched corymb is called **Compound corymb**. Example: Cauliflower.

b. Umbel: An inflorescence with indeterminate central axis and pedicellate flowers arise from a common point of peduncle at the apex. Example: *Allium cepa*. **Compound umbel**: It is a branched umbel. Each smaller unit is called **umbellule**.Example: *Daucas carota*, *Coriandrum sativum*.

3. Main axis flattened:

The main axis of inflorescence is mostly flattened (convex or concav) or globose. A **head** or **capitulum** is determinate or indeterminate, group of sessile or sub sessile flowers arising on a receptacle, often subtended by an involucre.

a. Head: A head is a characteristic inflorescence of Asteraceae and is also found in some members of Rubiaceae and Mimosaceae.

Torus contains two types of florets: 1. Disc floret or tubular floret. 2. Ray floret or ligulate floret. Based on the type of florets present, the heads are classified into two types.

i. Homogamous head: This type of inflorescence exhibits single kind of florets. Inflorescence has disc florets alone. Example: *Vernonia,* Ray florets alone. Example: *Launaea*.

ii. Heterogamous head: The inflorescence possesses both types of florets. Example: *Helianthus, Tridax*.

Disc florets at the centre of the head are tubular and bisexual, whereas the **ray florets** found at the margin of the head which are ligulate and pistilate (unisexual).

The flower and inflorescence are subtended by a lateral appendage called bract. In sunflower, you may notice that the whorl of bracts forms a cup like structure beneath mimicking the calyx. Such whorl of bracts is called involucre. A group of bracts present beneath the sub unit of inflorescence is known as Involucre.

II. Cymose inflorescence.

Central axis stops growing and ends in a flower, further growth is by means of axillary buds. Old flowers present at apex and young flowers at base

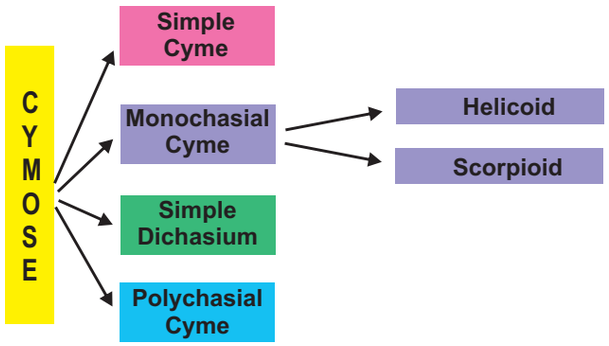
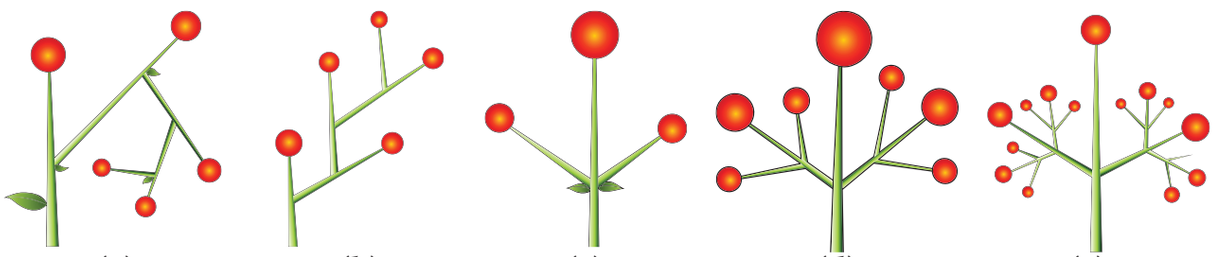


Figure 4.4: Cyme

1. Simple cyme (solitary): Determinate inflorescence consists of a single flower. It may be terminal or axillary. Example: terminal in *Trillium grandiflorum* and axillary in *Hibiscus*.

2. Monochasial Cyme (uniparous): The main axis ends with a flower. From two lateral bracts, only one branch grows further. It may be **Helicoid** or **Scorpioid.**



(a) **(b)** **(c)** **(d)** **(e)**

Figure 4.5: Diagrammatic representation of **(a)** Helicoid, **(b)** Scorpioid, **(c)** Simple dichasium, **(d)** Compound dichasium, **(e)** Polychasial Cyme

XIth_BIO-BOTANY_Ch_4-EM.indd 68 5/17/2020 11:14:34 PM